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Chapter 16

Water for Extraction—Composition, Recommendations, and Treatment

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1. INTRODUCTION

Water has received renewed attention over the last few decades in connection with many points along the coffee value chain. Areas of attention not only include farming and processing methods (Chapters 3 and 4), but also how extraction may affect the sensory properties of the coffee in the cup. Every barista knows that the choice of water either can highlight the specificities of a coffee, or may make it flat and dull. Historically, analysis of water quality has largely focused on safety and technical aspects. This is also reflected by the fact that the most common classification for water is based on hardness, a measure of the maximum amount of scale formation in coffee machine boilers or kettles.

The goal of this chapter is to equip the reader with a basic understanding of the influence water has on coffee and how to change the composition of water to alter its total hardness and alkalinity. [Section 2](#) will start by briefly introducing the basics of the science of water, which apart from in certain particular forms, such as distilled water or after treatment by reverse osmosis, contains dissolved compounds, such as minerals, gases, and organic molecules. Afterward, in [Section 3](#), we will explain the most important aspects of water composition, with a special focus on total hardness and alkalinity—both concepts are central to understanding the interaction between water and coffee. Common technical considerations with regard to scale formation and corrosion, which are a direct consequence of dissolved minerals in water, are also explained. [Section 4](#) provides an overview of the impact of water composition on technical and sensory aspects of coffee extraction. In this section we will also discuss current recommendations for “ideal” water composition and address flavor defects that

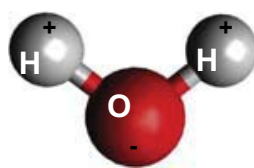


FIGURE 16.1 Structure of a water molecule and its charge distribution.

are caused by unsuitable water compositions. Since water, as we encounter it in its natural form, often does not correspond to what we consider the “ideal” composition for coffee preparation, total hardness and alkalinity first need to be adjusted, before the water is used for coffee extraction. [Section 5](#), therefore, will review the effect of water treatment methods on total hardness and alkalinity by introducing a systematic and novel approach that we believe is particularly useful and suited to the specialty coffee community. We will conclude this chapter by presenting practical examples on the impact of different water compositions in a cupping experiment and espresso extraction.

2. PHYSICOCHEMICAL CHARACTERISTICS OF WATER

Water is a molecule that is composed of one oxygen and two hydrogen atoms (H_2O) that are differently charged, leading to an uneven distribution of the electrical charge (see [Fig. 16.1](#)), known as polarity.

Due to its polar nature, water is a very good solvent for polar compounds, as is the case during coffee extraction. In contrast, it hardly dissolves any nonpolar compounds such as oils and fats, which make up more than 10% of the weight of roasted coffee. However, under high pressure and/or high temperature extraction conditions, such as espresso and moka pot, or alternatively over long contact times, such as French press, the solubility of nonpolar compounds is significantly increased ([Gloess et al., 2013](#)). The polarity of water is also responsible for a phenomenon called self-dissociation of water, in which a proton is transferred from one water molecule to another. This naturally leads us to the pH, which is calculated as the negative logarithm of the proton concentration,¹ where the higher the concentration of protons the lower the pH. For example, pH 6 corresponds to a concentration of protons in water of 10^{-6} mol/L. For every increase in the pH value of one unit (+1) the concentration of protons decreases by a factor of 10 and vice versa. The pH of pure water is 7, although this is only strictly true at 25°C.

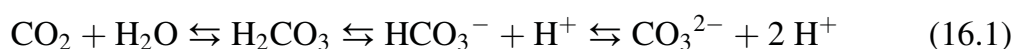
1. The pH is a scale from 0 to 14 that defines how acidic a solution is. More acidic solutions have lower pH. More alkaline solutions have higher pH. For example, coffee has a pH somewhere around 5.5 and is hence slightly acidic. To find out what the pH of a solution is, the concentration of protons (H^+) in the solution must be measured. The formula for calculating pH: $\text{pH} = -\log c(\text{H}^+) \cdot c(\text{H}^+)$ is the concentration of protons in the solution as mol per liter (mol/L). For a definition of mol, please consult a chemistry textbook. In essence mol/L tells us how much of the respective compound is present in 1 L of the solution. Once you know the concentration of H^+ in the solution, the negative value of the logarithm of the concentration of H^+ provides the pH value.

2.1 Carbon Dioxide and Carbonic Acid

Acids are key to coffee quality; specific acids can convey a fresh and lively sensation to the cup, a characteristic of some renowned specialty coffee origins. Chemically, an acid is a substance capable of donating a proton. An acid of particular importance to water and coffee is carbonic acid (H_2CO_3). It plays a central role in understanding various properties of water in relation to coffee. This section lays the foundation for understanding how the different carbonate species are related to each other; the applied aspects of which will be explored later in [Sections 3.4, 4, and 6.2](#).

Water naturally dissolves carbon dioxide (CO_2) from the air and roasted coffee provides another important source of CO_2 . During roasting CO_2 is generated (up to 2% of the weight of freshly roasted coffee is entrapped CO_2), which is subsequently released during extraction. Carbon dioxide dissolved in water forms carbonic acid (H_2CO_3), a weak acid that dissociates to hydrogen carbonate (HCO_3^-) plus one proton (H^+). The proton released into the water makes the water more acidic and the pH is reduced. At natural concentrations of CO_2 in the air, water in equilibrium with the air is slightly acidic with a pH of 5.7.

The hydrogen carbonate (HCO_3^-) still has a second proton that it could release; however, this only happens at pH values above 8.3, resulting in a doubly charged carbonate ion (CO_3^{2-}). [Eq. \(16.1\)](#) summarizes the chain of chemical reactions that extends from carbon dioxide dissolved in water (left) through to a carbonate ion that has released two protons (right):



3. WATER COMPOSITION

In practice, tap water from a lake or re-infiltrated from a river is on average significantly lower in mineral content than groundwater that has a typical residence time in the order of months to years. Furthermore, areas with bedrock made of carbonates are higher in mineral content than areas of silicate rock. For instance, in Switzerland the mineral content of the groundwater in the Central Plateau (fine sediments rich in carbonates) is about five times higher than in the central and southern Alps (crystalline rock dissolves more slowly and contains little carbonates).

3.1 Alkalinity

Now that we have introduced some basic properties of water, we can address two central concepts linking water to coffee that will guide our discussion—alkalinity and total hardness. Alkalinity represents the capacity of water to buffer acids. The amount of acid that has to be added to a water sample, or even to a coffee brew, until a specific pH is reached is a measure of its

alkalinity. If, for example, an acid is added to a coffee brew, its pH will decrease. The more acid that has to be added to reach a specific pH, the higher the alkalinity. Alkalinity should not be confused with a solution that has an alkaline pH, which simply means that its pH is higher than 7 (at 25°C). Alkalinity hence reflects the capacity of water or coffee to resist a change in pH, when acids are added, irrespective of its actual pH value. For freshwater, alkalinity can be calculated by the following equation:

$$\text{Alkalinity} = \text{HCO}_3^- + 2 \times \text{CO}_3^{2-} + \text{OH}^- - \text{H}^+ \quad (16.2)$$

—H⁺ (expressed as concentrations, mol/L)

For water with a pH below 8.3, alkalinity can be accurately approximated from the HCO_3^- content [Standard Methods for the Examination of Water and Wastewater (SMWW), 2012]. This is because all other components (CO_3^{2-} , OH^- , and H^+) are present in concentrations several orders of magnitude lower than that of HCO_3^- . Above pH 8.3, a substantial contribution from the carbonate ion (CO_3^{2-}) must also be considered, though this only occurs in regions with very hard water (more than 370 ppm CaCO_3 of alkalinity). Although this is rare, it occurs in some regions that are characterized by dominant carbonate bed rock. For example in the region of the canton of Zurich, at least 10% of all tap water has an alkalinity above 370 ppm CaCO_3 of alkalinity.

Alkalinity is also very important in relation to its effect on sensory properties; it is the opposite of acidity in chemistry as well as in coffee sensory terminology (see Section 4.2). Additionally, and most importantly, alkalinity can be measured by commercially available test kits. Test kits measure the amount of acid that has to be added to reach a specific pH (most commonly pH 4.5), by counting the number of standard drops that need to be added until the color of the water changes. It should be noted that these test kits are often incorrectly marketed as carbonate hardness test kits—see next section for the distinction.

3.2 Hardness of Water

In this chapter the term “hardness” will be used to denote “total hardness,” which is defined in accordance with worldwide industrial norms (SMWW; DIN 38409-6, 1986; ASTM D-1126, 2002; EPA Method, 130.2, 1982) as the sum of amounts or equivalent concentrations of calcium and magnesium (see Section 3.3 for calculation). In contrast, “carbonate hardness” is defined by the maximum amount of scale that can form for a given water composition, and is determined by the common minimum of total hardness and alkalinity, whichever is lower. During scale formation both total hardness and alkalinity are reduced in equal amounts. Most natural water sources have a total hardness that is higher than their alkalinity, as depicted in Fig. 16.2A where the carbonate hardness is equal to the alkalinity. However, water treated with a so-

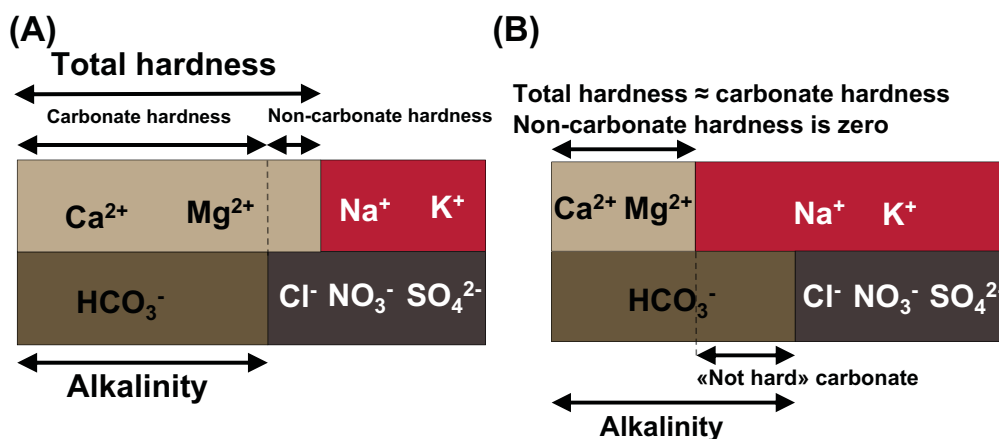


FIGURE 16.2 Terms used with respect to hardness and alkalinity for two sample water compositions—size is proportional to equivalent concentrations: (A) total hardness > alkalinity; (B) total hardness < alkalinity.

called softener that removes calcium or magnesium and replaces them with sodium or potassium (see Fig. 16.2B) has a reduced total hardness, without affecting the alkalinity. In this case, total hardness is lower than alkalinity and carbonate hardness is equal to total hardness.

3.3 Units of Hardness

When calculating the hardness of water, it is crucial to distinguish between amount concentrations, such as mol/L (where 1 mol can be understood to be the “chemical dozen” equaling 6.4×10^{23} atoms, ions, or molecules), and mass concentration (e.g., mg/L on bottled waters). Although amount concentration refers to the number of entities in a given volume of water, mass concentration refers to the weight of compounds in the same water volume. Mass concentration does not reflect the true proportions of different substances and is, therefore, unsuitable for comparing the amount of different compounds in a water sample. This can best be illustrated by a simple example: When considering the compounds that contribute to the total hardness of water, 1 mmol/L of Ca^{2+} is equal to 40 mg/L while 1 mmol/L of Mg^{2+} is equal to 24.3 mg/L. For the alkalinity of water, 1 mmol/L of HCO_3^- is equal to 61 mg/L. In all three cases the amount concentrations and hence the number of compounds are equal (namely 1 mmol/L), whereas the mass concentrations are obviously different, as the compounds all have different weights.

From a chemist’s perspective, amount concentrations such as mol/L should be used since the values are proportional to the actual number of atoms, ions, or molecules in a specific volume of water.

It is also important to note that, throughout industry, hardness values are referred to in equivalent units, such as ppm CaCO_3 (USA) or German ($^\circ\text{d}$) degrees. Compared to amount concentration (such as mol/L) equivalent units,

the molar concentrations of Ca^{2+} , Mg^{2+} , and HCO_3^- scale effectively as they react together. Therefore, the equivalent concentrations of calcium and hydrogen carbonate always decrease by the same amount when scale forms or during specific water treatment methods (see [Fig. 16.5](#)).

[Table 16.1](#) provides an overview on the conversion factors for the most commonly used units in water analysis; the values have been calculated based on the [IUPAC Periodic Table of the Elements \(2013\)](#) and are also in agreement with [Hem \(1985\)](#) and the DIN norm for water hardness [\(1986\)](#).

Please note that the standard unit used by the Specialty Coffee Association of America (SCAA) “ppm CaCO_3 ” (or alternatively “mg/L CaCO_3 ”) is often misleadingly abbreviated as ppm or mg/L for both total hardness and alkalinity (see [Section 4.3](#) on water standards), since “straight” mass concentrations (as labeled on water bottles) does not correspond to the true proportions of calcium, magnesium, and hydrogen carbonate.

3.4 Scale Formation and Corrosion

Water boilers in the home and industrial equipment can become encrusted with scale residues that are mainly composed of calcium carbonate (CaCO_3) and at high pH (>10) also of magnesium hydroxide ($\text{Mg}(\text{OH})_2$). The latter occurs in steam boilers, especially with sodium-softened water. The speed of this process is determined by the solubility of the calcium carbonate for the given temperature and pH conditions. The primary technical concerns in terms of scale formation when using hard water are the decrease in efficiency of the heating system (when the layer of scale acts as an insulator) and the blockage of valves and flow restrictors (usually called gicleurs in coffee machines). On the other hand, water that is very low in alkalinity and can easily become acidic (since insufficient acid buffer is present), which can cause corrosion of metal parts. To increase longevity and hence reduce maintenance costs for coffee machines, a number of countries have issued recommendations to minimize the costs of scale formation (high hardness and alkalinity values) and corrosion (generally low mineral content and low alkalinity in particular). In Switzerland, there is only a recommendation with regard to hardness that specifies the optimal hardness as 12–15°fH ([SVGW, 2008](#)). In other countries, optimal ranges for alkalinity are provided that are typically between 5 and 6°fH ([Navarini and Rivetti, 2010](#)). As an additional measure, coffee machine companies recommend regular descaling using acid dissolved in water (preferably one with little taste), and also offer commercial solutions.

4. IMPACT OF WATER COMPOSITION ON EXTRACTION

Water that has no minerals other than calcium or magnesium carbonate (CaCO_3 , MgCO_3) contains equal amounts of hardness and alkalinity (in hardness equivalent units). A total hardness that is much higher than alkalinity indicates that there is a significant amount of sulfate present in the water.

TABLE 16.1 Conversion Factors for Units of Hardness and Alkalinity—Rounded to Four Significant Digits—Given in Bold Are the Most Commonly Used Conversion Factors											
		ppm CaCO ₃	°dH	°fH	gpg	°e	Ca ²⁺ + Mg ²⁺ (mmol/L)	HCO ₃ [−] (mmol/L)	Ca ²⁺ (mg/L)	Mg ²⁺ (mg/L)	HCO ₃ [−] (mg/L)
ppm CaCO ₃ (=mg CaCO ₃ /L)	1 ppm CaCO ₃ =	1	0.05603	0.1	0.05842	0.07022	0.009991	0.01998	0.4004	0.2428	1.219
German degrees (°dH)	1°dH =	17.85	1	1.785	1.0423	1.253	0.1783	0.3567	7.147	4.334	21.76
French degrees (°fH)	1°fH =	10.00	0.5603	1	0.5842	0.7022	0.09991	0.1998	4.004	2.428	12.19
Grains per US gallon (gpg)	1 gpg =	17.12	0.9591	1.712	1	1.202	0.1710	0.3421	6.855	4.157	20.87
English degree (°e)	1°e =	14.24	0.7979	1.424	0.8320	1	0.1423	0.2846	5.703	3.458	17.36
Ca ²⁺ + Mg ²⁺ (mmol/L)	1 mmol/L =	100.1	5.608	10.01	5.847	7.028	1	—	40.08	24.30	—
HCO ₃ [−] (mmol/	1 mmol/L =	50.04	2.804	5.004	2.923	3.514	—	1	—	—	61.02
Ca ²⁺ (mg/L)	1 mg/L =	2.497	0.1399	0.2497	0.1459	0.1753	0.02495	—	1	—	—
Mg ²⁺ (mg/L)	1 mg/L =	4.118	0.2307	0.4118	0.2406	0.2891	0.04114	—	—	1	—
HCO ₃ [−] (mg/L)	1 mg/L =	0.8202	0.04595	0.08202	0.04791	0.05759	—	0.01639	—	—	1

Water and its minerals are arguably the most important ingredients in the quality of a coffee beverage after the roasted and ground coffee itself (Navarini and Rivetti, 2010). Due to its potential for scaling, water is commonly treated to reduce its hardness to avoid the precipitation of minerals. A further factor that is equally important as water composition in terms of its impact on extraction is the mass ratio of water to roasted and ground coffee, since this can differ by as much as a factor of 10 for different extraction methods. An extreme example is an espresso, which can be prepared with a dry coffee mass to beverage mass ratio of 1:2 (16 g of coffee to make two cups, each containing a beverage of 16 g or less). On the other hand, a typical ratio for drip coffee is 1:15 (i.e., 60 g of coffee per 1000 g of water resulting in approximately 880 g of filter coffee). Therefore, the extent of the acid buffering effect of the alkalinity (of the water used for brewing) is much lower in an espresso than in a filter coffee—see Section 4.1. The extraction pressure must also be considered when assessing the extraction of coffee with water, since the solubility of carbon dioxide increases drastically with pressure (Sánchez et al., 2016). There is mounting evidence (Hendon et al., 2014) that calcium and magnesium are major contributors to the efficiency of coffee extraction. Moreover, it has been suggested that magnesium (per hardness equivalent or mole) is more efficient in extracting constituents from coffee (Hendon et al., 2014). Although the study of the impact of water composition on filter coffee dates back to 1950, espresso has only been under investigation in the last 20 years. In the studies presented below three different situations are described:

1. At high alkalinity (>100 ppm), more of the acids extracted from coffee are neutralized by hydrogen carbonate, thereby producing carbonic acid that can degas as carbon dioxide, depending on the pressure and temperature. The generation of carbon dioxide during extraction creates additional resistance and thereby prolongs the extraction time under otherwise equal circumstances (Fond, 1995; Rivetti et al., 2001). See Section 6.2 for an applied example of espresso extraction with water treated by an ion exchanger that changes either total hardness or alkalinity.
2. High alkalinity and concurrent low hardness lead to increased pH when water is heated—this effect is even more pronounced in the presence of sodium. It has been suggested to be the cause of increased resistance during extraction due to decreased solubility of carbohydrates in water at high pH (Navarini and Rivetti, 2010).
3. At high hardness and equally high (or lower) alkalinity, no increase in percolation time was observed (Fond, 1995; Rivetti et al., 2001).

4.1 Buffering of Coffee Acidity

Perception of acidity in coffee has been studied extensively (for an overview see Gloess et al., 2013). The most commonly cited phenomenon is the correlation of

sensory perception of acidity with titrable acidity of coffee extracts (titration to a pH of 6.6, which corresponds to mouth pH). From a chemical point of view this would also make sense as stated by Clifford (1988): “In effect, the reaction of the acid with the receptor is a titration, and thus very similar to the process used in measuring titratable acidity.” Because alkalinity of water reacts with the extracted acids from coffee, it effectively neutralizes part of the acids. Based on data from Gloess et al. (2013) water can amount to as much as either 30% or 11% of the titrable acidity measured in filter or lungo espresso extractions, respectively (Gloess et al., 2013; authors’ measurements and calculations). These percentages were measured for water with an alkalinity of 50 ppm CaCO_3 (1 mmol/L) that was used to extract a beverage from a medium to dark roasted coffee (Guatemala, Antigua, la Ceiba, 80 Pt on Colorette 3b, ~45 Pt on Agtron M-Basic scale). As a result of Clifford’s suggestion, a simple equation can be formulated for perceived acidity, where the perceived acidity of coffee equals the acidity extracted from the coffee diminished by the alkalinity of the water.

4.2 Ideal Water Composition for Coffee Extraction

The most common standard in the coffee industry with respect to sensory properties is the “SCAA Standard: Water for Brewing Specialty Coffee” (2009). It defines an optimum extraction at a total hardness of 68 ppm CaCO_3 (with an acceptable range of 17–85 ppm CaCO_3), an alkalinity of 40 ppm CaCO_3 , and pH 7 (acceptable range of pH 6.5–7.5). This area is indicated with a red line in Fig. 16.3, on which the optimum is marked with a circle (pH is not considered in the graph). In addition, it provides a target value of 10 mg/L for sodium, although sodium only becomes perceptible at concentrations above 250 mg/L (Pohling, 2015). The statement on “total chlorine,” which should be zero only refers to chlorine gas and hypochlorite (OCl^-), which are used as disinfectants and impart an unpleasant flavor. Tasteless ion chloride (Cl^-) on the other hand is not restricted. An optimum total dissolved solids (TDS) value of 150 mg/L is stated; however, the standard method of determining TDS using a conductivity meter has an uncertainty of approximately 30%. This uncertainty is primarily caused by a varying conversion factor for the transformation of the effective measurement of electrical conductivity from $\mu\text{S}/\text{cm}$ to TDS in mg/L, which, depending on the mineral composition, can vary between 0.5 and 1.0. Additionally, most so-called TDS-meters do not measure and take into account water temperature, which can lead for example, to a further 20% variation in the measurement for a 10°C change in temperature (e.g., tap temperature to room temperature). Furthermore, it is not clear which anions are suggested to achieve charge neutrality. Fig. 16.3 summarizes the recommended water compositions for coffee extraction according to the SCAA, Colonna-Dashwood and Hendon (2015, given in ppm CaCO_3 without transformation), Rao (2008), and Leeb and Rogalla (2006). The figure also contains descriptors illustrating taste

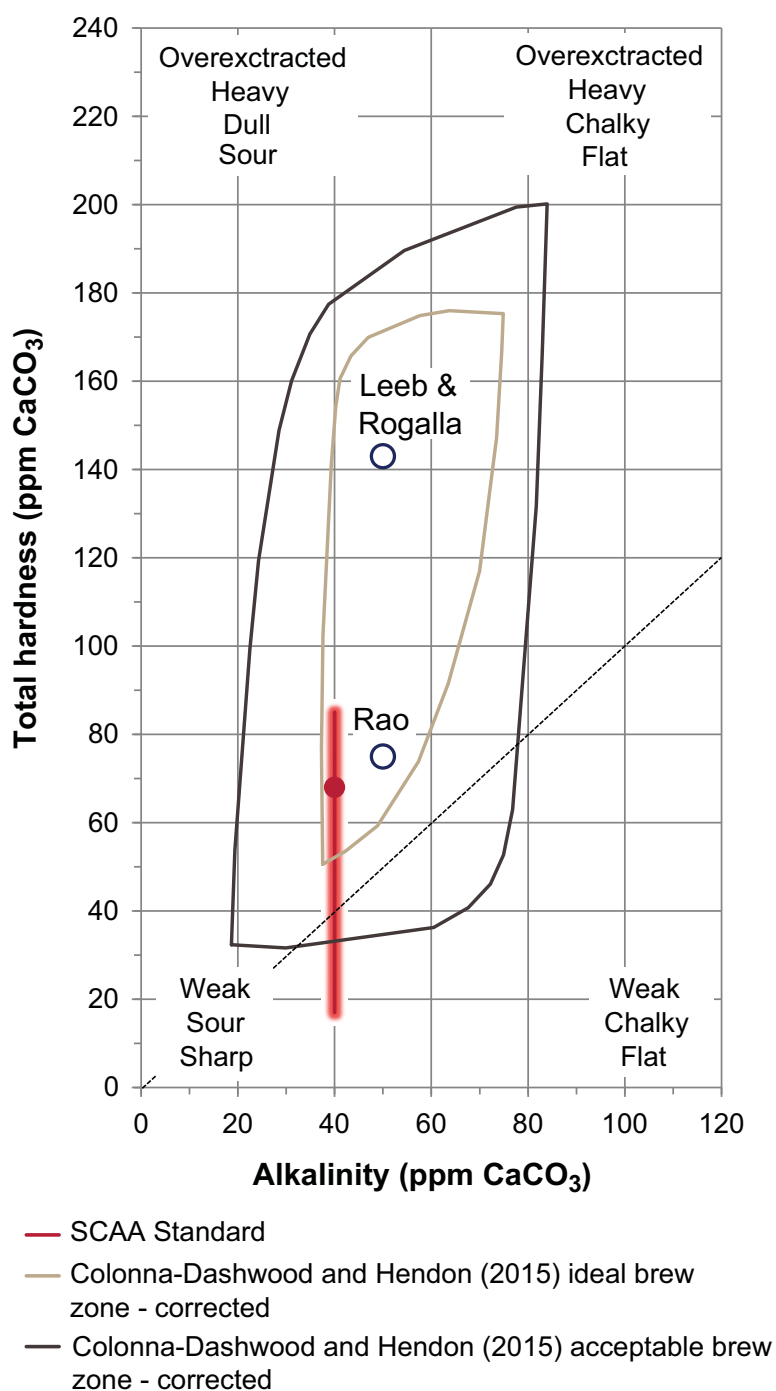


FIGURE 16.3 Recommended water compositions.

imbalances or off-notes that arise from suboptimal values for either total hardness (impact on extraction efficiency) or alkalinity (impact on degree of buffering of the coffee acids) based on reports from [Rao \(2013\)](#) and [Colonna-Dashwood and Hendon \(2015\)](#). As explained in [Section 3.4](#), a deviation from the diagonal line indicates the presence of ions other than magnesium, calcium, and hydrogen carbonate. For water with a total hardness higher than its alkalinity, as in most recommendations, this surplus is associated with the presence of chloride or sulfate.

5. WATER TREATMENT

In this section, we will present a novel and practical way of describing water treatment methods in terms of hardness and alkalinity. In general, water treatment methods can be classified into five common categories:

- **Filtration:** Removal of particles, microbes, or organic compounds responsible for off-flavors.
- **Ion exchanger:** Cation exchange [e.g., magnesium and calcium ions are exchanged for hydrogen ions (protons) or sodium ions]—combined cation, and anion exchangers produce deionized, almost pure H_2O , which is then mixed again with tap water to adjust its mineral content.
- **Reverse osmosis:** Nonselective removal of all dissolved solids by filtration through a semipermeable membrane, i.e., only permeable for water but not for other components present in water.

Other methods that can even further purify water are distillation and precipitation. However, these methods are expensive and/or elaborate and are not commonly used for water for coffee brewing.

With respect to coffee extraction, odor free, clean drinking water can be accurately characterized by its hardness (as a measure of extraction efficiency) and alkalinity (as a measure of its acid buffering capacity). [Fig. 16.4](#) provides an overview of the changes in hardness and alkalinity resultant from the most common treatment methods, using two different initial water compositions [1](#) and [2](#):

- **a**: Softener—Cation exchanger: Ca^{2+} and Mg^{2+} for potassium (K^+) or sodium (Na^+) only affects hardness and is, therefore, vertically oriented in [Fig. 16.4](#). The alkalinity is not affected.
- **b**: Decarbonizer—Cation exchanger: Ca^{2+} and Mg^{2+} for H^+ —oriented diagonally with a slope of 1. The net effect is that the change in alkalinity equals the change in hardness. See [Section 6.2](#) for a discussion of the effect of this treatment on espresso extraction.
- **b***: Combination of decarbonizer (**b**-type) with a small fraction of softener (**a**-type).
- **c**: Demineralizer: Reverse osmosis (RO) or deionizer by ion exchange—removing ions irrespective of the initial composition; RO can also be used to increase the mineral content by mixing concentrated water with untreated water before the membrane—oriented toward the point of origin (0/0) or away from it.
- **d**: Dealkalizer: Anion exchanger: HCO_3^- for Cl^- (not yet commercially available for coffee applications) or addition of a strong acid (e.g., HCl). This will lead to a reduction of the alkalinity, without affecting its total hardness.
- **e**: not shown: Cation exchange of Ca^{2+} for Mg^{2+} —does not change either hardness or alkalinity, so both values remain constant. However, the mineral composition of the water will be altered.

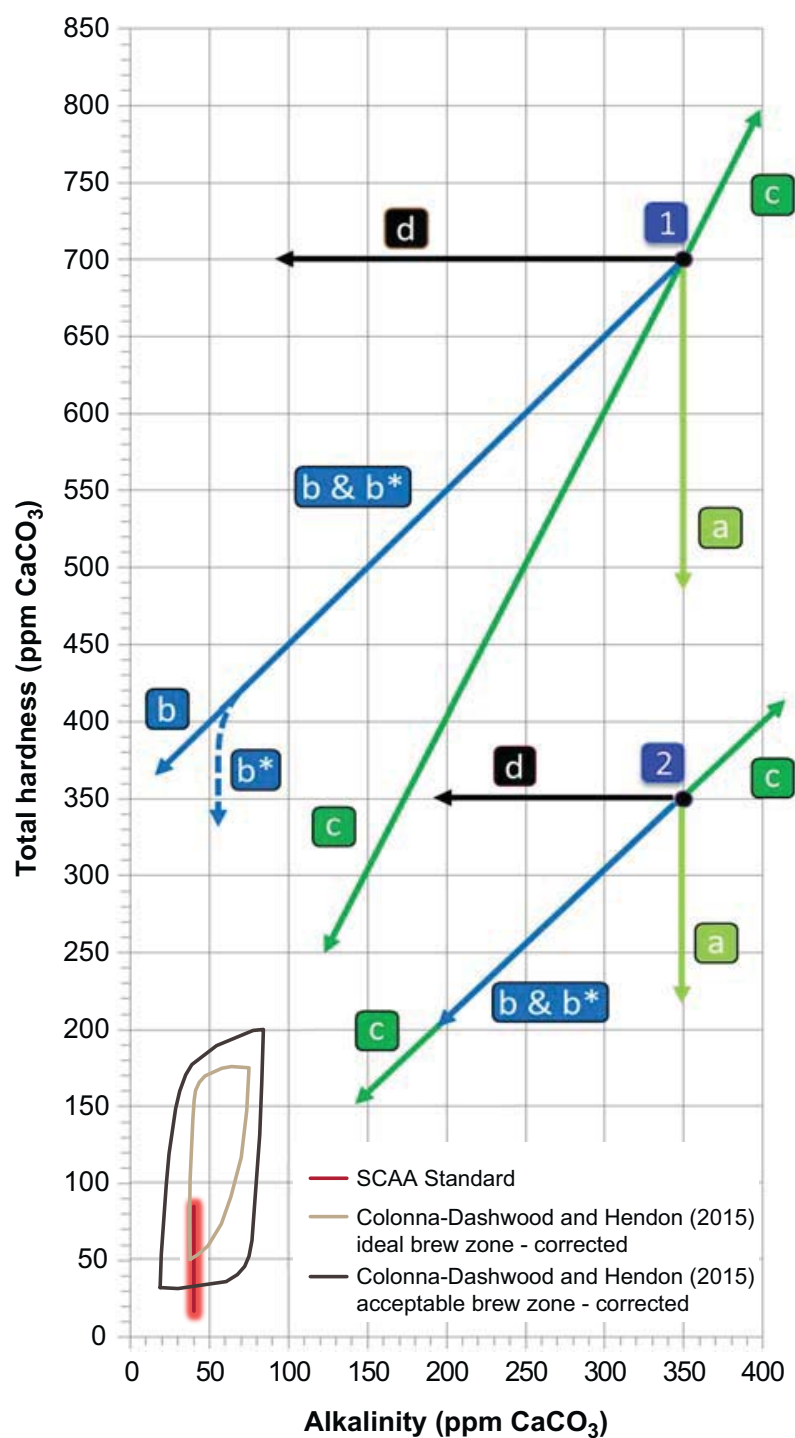


FIGURE 16.4 Changes in hardness and alkalinity for five treatments and two initial compositions (**1** and **2**). **a**: Cation exchanger: $\text{Ca}^{2+} + \text{Mg}^{2+}$ versus K^+ or Na^+ . **b**: Cation exchanger: $\text{Ca}^{2+} + \text{Mg}^{2+}$ versus H^+ . **b***: Mixture of mostly b-type and some a-type ion exchanger. **c**: Demineralizer; **d**: anion exchanger: HCO_3^- versus Cl^- .

6. APPLIED EXAMPLES

In this section, we explore how the contents of the previous section can be combined to investigate the effect of different water compositions on extraction. This involves the determination of the starting water composition, the choice of a target water composition, and a suitable water treatment method for achieving the desired change in composition.

6.1 Cupping With Different Waters

To investigate the impact of water composition on the sensory properties of coffee extraction, a cupping experiment was conducted ([Wellinger and Yernetzian, 2015](#)). This experiment involved three different water compositions prepared by using a decarbonizer-type ion exchanger that reduces total hardness and alkalinity in equal amounts. The water compositions obtained are shown in [Fig. 16.5](#) and the alkalinity is varied by $\pm 50\%$ relative to the SCAA target of 40 ppm CaCO_3 alkalinity. For reference, the water compositions as used in the World Barista Championship (WBC) 2013 in Melbourne and 2014 in Rimini are also depicted (data from [Colonna-Dashwood and Hendon, 2015](#)).

The experiment used the following two coffees:

1. Colombia, washed process, Caturra and Castillo, screen size 15, La Argelia farm, Tolima region, 1580–1900 m.
2. Brazil, natural process, yellow bourbon, Tupi, Icatu, and Yellow catuai, screen size 16–18, Lagoa Formosa farm, Minas Gerais region, 1000–1200 m.

The coffees were prepared in a blind tasting session according to the SCAA cupping protocol with slight modifications to the sensory attributes: 13.8 g of freshly ground coffee was used with 250 mL water at 93°C. Fragrance, uniformity, and clean cup scores were not evaluated. Sweetness was evaluated on a scale of 1–10. [Fig. 16.6](#) shows the average scores computed from three repetitions and three cuppers: either certified Q grader or WBC judge. When significant differences were observed, the highest score was attributed to the coffee samples brewed using low or medium mineral content water. It was generally observed that the water with the lowest mineral content scored highest for most attributes in both coffees. Across the spectrum, the Colombian coffee scored higher than the Brazilian one. For both coffees, significant differences in the flavor and acidity attributes were observed ($\alpha = 5\%$), with the lowest hardness and alkalinity scoring the highest. Increased hardness and alkalinity resulted in gradually lower scores. Additionally, the Colombian coffee exhibited significant differences in the scores for balance and overall, whereas the scores for after-taste were significantly different for the Brazilian coffee.

The results show that even relatively small changes in hardness and alkalinity significantly affect the sensory attributes of a coffee. Although

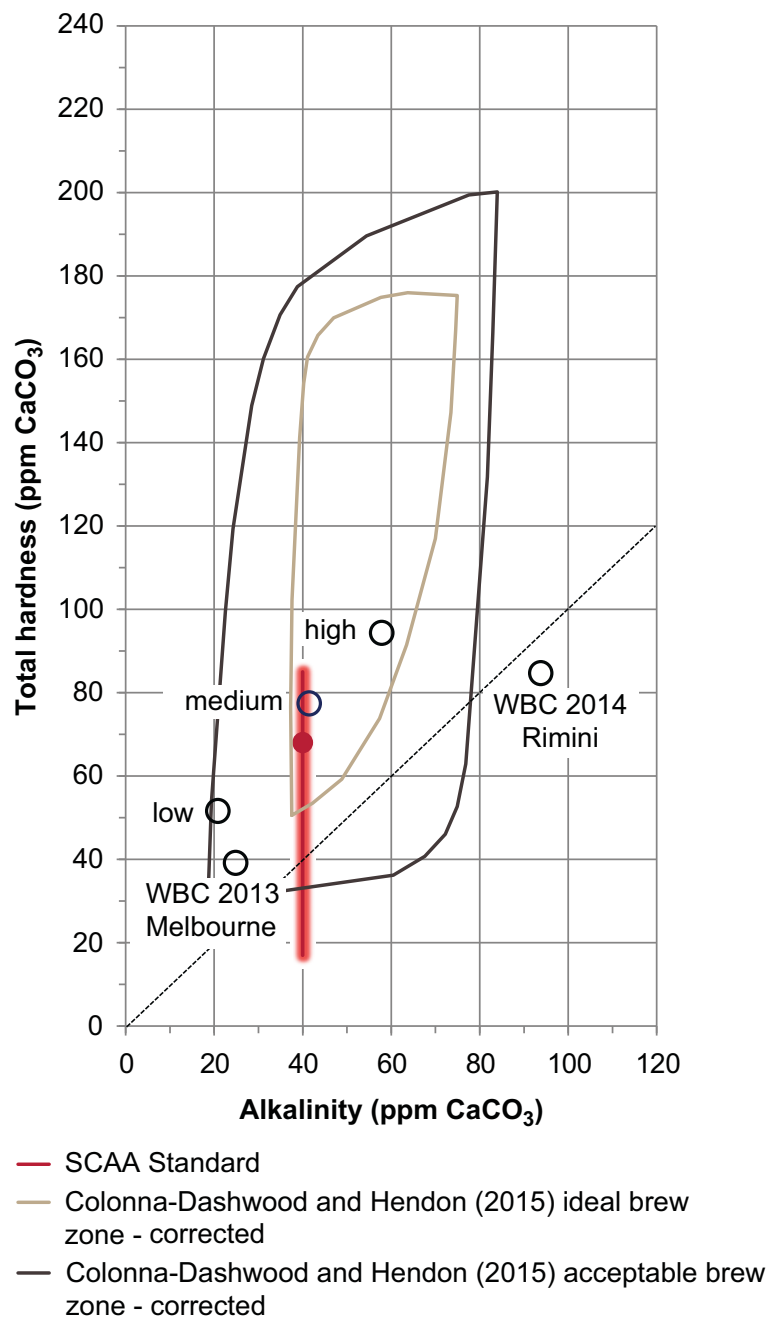


FIGURE 16.5 Water compositions used compared to existing recommendations and WBC competitions water.

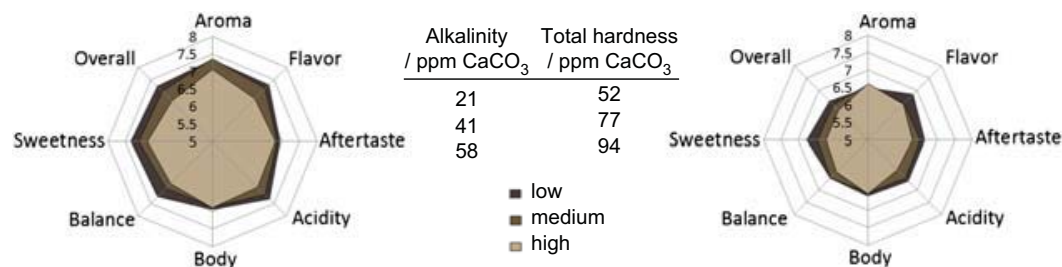


FIGURE 16.6 Cupping scores of two different coffees prepared with water containing three different levels of hardness and alkalinity: low, medium, and high (Wellinger and Yeretizian, 2015).

further research is needed to clarify the mechanisms that cause these attribute changes, we suspect that this is the combination of two effects: (1) the total hardness affecting the extraction efficiency and (2) the alkalinity affecting the perceived acidity. In summary, the results demonstrate that although general tendencies can be formulated for how a specific change in hardness or alkalinity impacts the sensory attributes of a coffee beverage, there are differences between different coffees that are closely linked to their overall flavor profile. The fact that the low mineral content water received the highest score for most attributes could be related to the relatively high extraction yield that is typically achieved by the cupping method (attributed to extended contact time compared to other brewing methods), which also leads to the perception of heavier body compared to drip coffee for instance. Hence, the results shown here might well be quite different if repeated with the same coffees but another extraction method, such as filter.

6.2 Water Decarbonization and Espresso Extraction

A common problem encountered when using a decarbonizer, exchanging Ca^{2+} and Mg^{2+} for H^+ , to treat hard water for use in espresso machines is the formation of carbonic acid, which leads to excess dissolved carbon dioxide in the water. The protons that are released, in exchange for Ca^{2+} and Mg^{2+} , neutralize the hydrogen carbonate present in the water forming carbonic acid that can in turn degas as carbon dioxide, depending on the pressure and temperature. In a café where the water from the ion exchange cartridge stays pressurized all the way up to the espresso machine, the carbonic acid cannot escape as carbon dioxide and subsequently can significantly impact the extraction. Carbon dioxide becomes more soluble as pressure increases and less soluble as temperature increases. When the water containing high amounts of carbonic acid enters the extraction chamber (basket) it creates an additional resistance (see [Section 4](#)) in an identical way to the carbon dioxide content from the freshly roasted and ground coffee. Therefore, when hard water is treated with a decarbonizer, a similar effect may be observed in espresso extraction as observed for coffee that has been freshly roasted before extraction (typically <3 days postroast). Freshly roasted coffee forms a large amount of crema with untypically large bubbles that collapse much quicker than those formed from the same coffee that has been rested for a longer period. For example, a reduction in the alkalinity by 200 ppm CaCO_3 alkalinity will increase dissolved carbon dioxide by 176 mg/L. Therefore, if we consider the extreme example of extracting a double espresso using 15 g of Arabica coffee that has been freshly ground (2 min) just 1 h after roasting using 30 g of decarbonized water, the water would add another 20% to the carbon dioxide already contained in the coffee grounds. The excess carbon dioxide produced by decarbonization could be removed by a process called degassing, which is well established in other industrial applications. Alternatively the water could

be treated with a demineralizer that removes all ions without generating additional dissolved carbon dioxide.

7. CONCLUSIONS AND OUTLOOK

Given clean water that is free of particulates and off-flavors, a water sample can be accurately characterized for its use in coffee extraction by the measurement of hardness and alkalinity; both can be measured by titrations, for which simple to use commercial kits exist. Although research and industry recommendations for hardness levels propose a relatively wide ideal range (as much as a factor of 5), the recommended variation for alkalinity is much smaller (factor of 2). As explained in [Section 5.3](#) the conductivity of water, as measured by a conductivity meter (sometimes incorrectly called a TDS-Meter), is not a meaningful parameter on its own, except for obtaining a rough estimate of the total solid content. However, for any given water source (e.g., tap water or bottled water) and a defined water treatment method, measuring the conductivity and correcting for temperature offers the opportunity to detect changes in the output water composition caused by changes in either the tap water composition or the efficiency of the water treatment method. Since conductivity can be measured with inexpensive online sensors, this offers the opportunity to perform continuous real-time monitoring of water quality.

The aim of this chapter has been to provide a scientific and quantitative framework to discuss water, its composition and its impact from the perspective of the coffee industry. This includes a description of the various water treatment methods used within the (specialty) coffee industry and how water quality impacts the extraction process as well as the sensory properties of the brew. In the past, water was often regarded as something exclusively relevant to technical issues, such as maintenance intervals and the like, rather than a relevant parameter for coffee extraction and the quality of the resulting cup. As shown in [Section 6.1](#) the sensory impact of different water compositions is also dependent on the specific type of coffee used. Therefore, the existence of a single optimum value for total hardness and alkalinity is most probably unlikely, because this optimum depends on the coffee (both its green quality and the roasting process applied), the extraction method, and last but not the least the sensory preferences of the targeted consumer group for a specific beverage. As water comes into focus as an important parameter for the sensory properties, this could also impact the way coffee is roasted. Many roasters might have chosen their style of roasting based on the specific water composition they have, since it would take a significant additional effort to test every roast with different waters ([Colonna-Dashwood and Hendon, 2015](#)).

To extend our database and experience, areas for future research should include applied studies on the impact of water composition on extraction efficiency and the sensory impact on espresso coffee. Since the ratios of water to coffee for espresso and for filter coffee are very different, much higher total

hardness and alkalinity (both over 100 ppm CaCO_3) might still result in a high-quality espresso beverage. Nevertheless, the use of such water will lead to scale formation and is, therefore, not recommended on a permanent or long-term basis. Moreover, the impact of water composition on flow speed and therefore contact time must also be considered when designing further experiments to determine the influence of water composition on the resulting cup. As demonstrated with the experiments in [Section 6.1](#), not every coffee will be affected identically by a specific change in water composition. In addition, the impact of, for example, pH or specific substances present in the water on aroma binding or volatility is an unexplored field in coffee. Therefore, further investigations on the effect of different parameters, such as total hardness and alkalinity, on different coffees and their sensory attributes represent a promising and important field of research. The often suggested differences between calcium and magnesium should also be further investigated—it is suggested that the impact of identical concentrations of each on the sensory profile differs. In case there are significant differences, we recommend reassessing the current standard for total hardness as a sum of calcium and magnesium to a new standard where the calcium and magnesium content are measured and considered individually to achieve optimum results in the cup.

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